



Universidade do Minho
Escola de Engenharia
Departamento de Informática

KNIME
Lego Dataset

MESTRADO EM ENGENHARIA DE TELECOMUNICAÇÕES E INFORMÁTICA
MESTRADO integrado EM ENGENHARIA DE TELECOMUNICAÇÕES E
INFORMÁTICA
Inteligência Artificial para as Telecomunicações
2025/2026

Objectives

- The primary goal of this task is to **create a comprehensive and efficient workflow** tailored specifically for pre-processing the Lego dataset.
- The final result will be used to train a **Linear Regression** model.

Dataset- Features

10 attributes

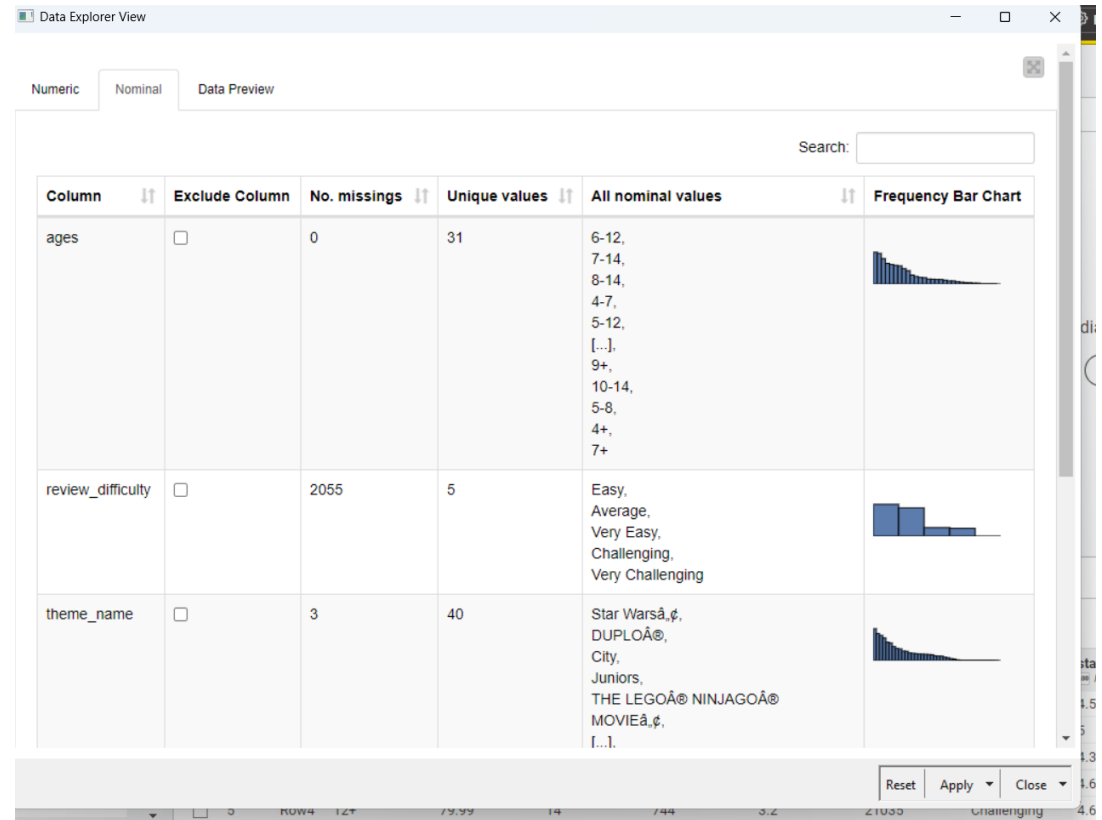
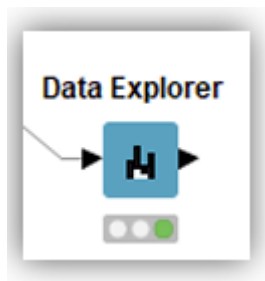
- **'age'**: What age category do you belong to
- **'list_price'**: set price (in \$)
- **'num_reviews'**: Number of revisions per set
- **'piece_count'**: Number of pieces in that LEGO set
- **'play_star_ratings'**: ratings
- **'review_difficulty'**: level of difficulty of the set
- **'star_rating'**: ratings
- **'theme_name'**: Which theme it belongs to
- **'val_star_rating'**: ratings
- **'country'**: Country name

Dataset- Exercise

- Launch Knime and create a **new workflow**.
- **Import dataset** with the correct **reader node**.
- Use the appropriate nodes for **data exploration**, such as noticing missing values, and a few basic statistics like variable distribution, minimum, max and mean scores.

Data Exploration

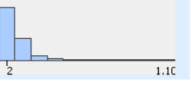
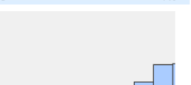
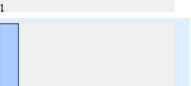
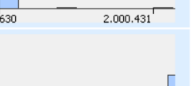
Offers a range of options for displaying properties of the input data in an interactive view.



Data Exploration

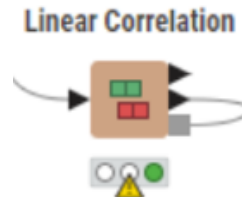
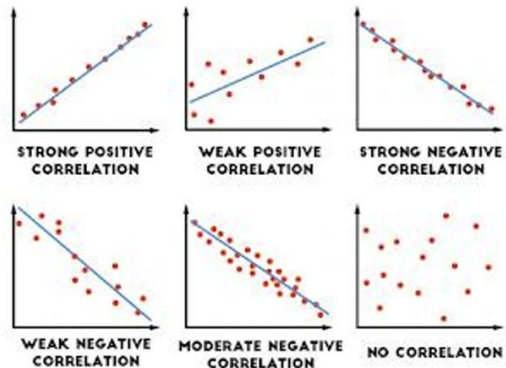


File

	Numeric	Nominal	Top/bottom										
Column	NUM	NUM	NUM	NUM	NUM	NUM	NUM	NUM	NUM	NUM	NUM	NUM	HISTOGRAM
list_price	2.2724	65.142	?	1,104.87	91.9804	4.6807	31.6525	0	0	0			
num_reviews	1	16.8262	?	367	36.369	5.2629	35.535	1,620	0	0			
piece_count	1	493.4059	?	7,541	825.3646	3.9681	19.8627	0	0	0			
play_star_rating	1	4.3376	?	5	0.6521	-1.6728	3.9361	1,775	0	0			
prod_id	630	59,836.7523	?	2,000,431	163,811.4523	11.4223	132.1878	0	0	0			
star_rating	1.8	4.5141	?	5	0.5189	-1.6045	3.3166	1,620	0	0			
val_star_rating	1	4.229	?	5	0.6603	-1.2171	1.9402	1,795	0	0			

Linear Correlation

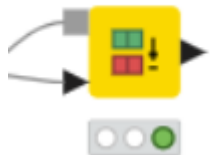
- **Correlation of -1:** perfect inverse correlation
- **Correlation of 0:** no correlation
- **Correlation of 1:** perfect correlation

[illegible]

Correlation Filter

It is used for **feature selection** based on **correlation**. It allows you to filter out features (columns) that have high correlation with one another.

Correlation Filter



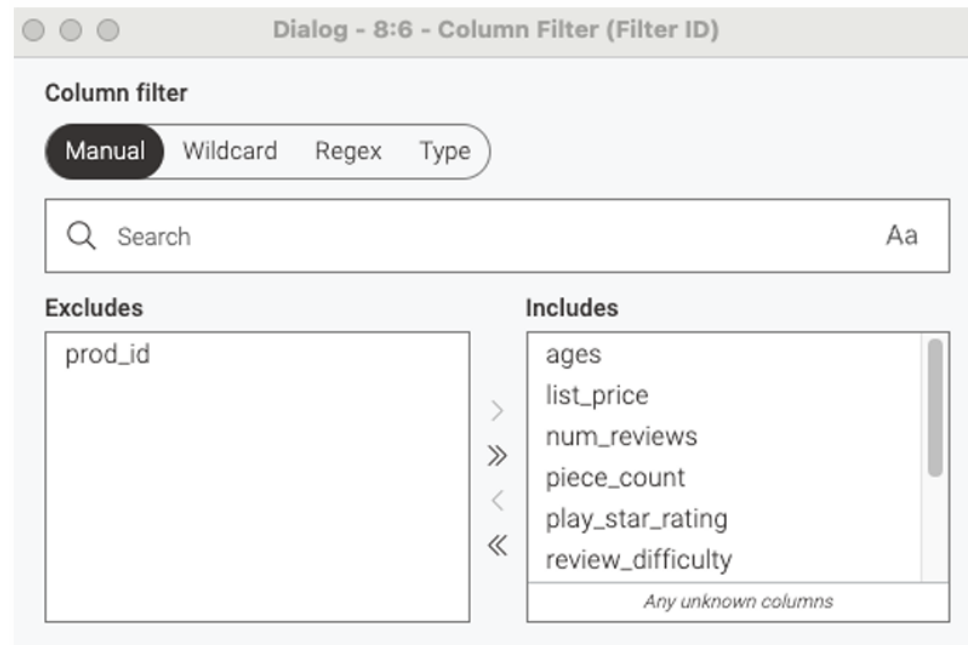
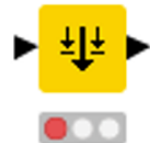
☐	#	RowID	ages Number (dou...	list_price Number (dou...	num_revi... Number (dou...	piece_co... Number (dou...	play_star... Number (dou...	prod_id Number (dou...	review_di... Number (dou...	star_rating Number (dou...	theme_ Number (
☐	1	ages	1	⊗	⊗	⊗	⊗	⊗	0.464	⊗	0.614
☐	2	list_p...	⊗	1	0.451	0.87	0.03	0.389	⊗	0.002	⊗
☐	3	num...	⊗	0.451	1	0.547	-0.038	-0.039	⊗	0.007	⊗
☐	4	piec...	⊗	0.87	0.547	1	0.022	0.218	⊗	0.074	⊗
☐	5	play_...	⊗	0.03	-0.038	0.022	1	-0.004	⊗	0.608	⊗
☐	6	prod...	⊗	0.389	-0.039	0.218	-0.004	1	⊗	-0.061	⊗
☐	7	revie...	0.464	⊗	⊗	⊗	⊗	⊗	1	⊗	0.39
☐	8	star_...	⊗	0.002	0.007	0.074	0.608	-0.061	⊗	1	⊗
☐	9	them...	0.614	⊗	⊗	⊗	⊗	⊗	0.39	⊗	1
☐	10	val_s...	⊗	-0.047	0.027	0.077	0.481	-0.082	⊗	0.728	⊗
☐	11	coun...	0.021	⊗	⊗	⊗	⊗	⊗	0.025	⊗	0.059

Remove unnecessary features

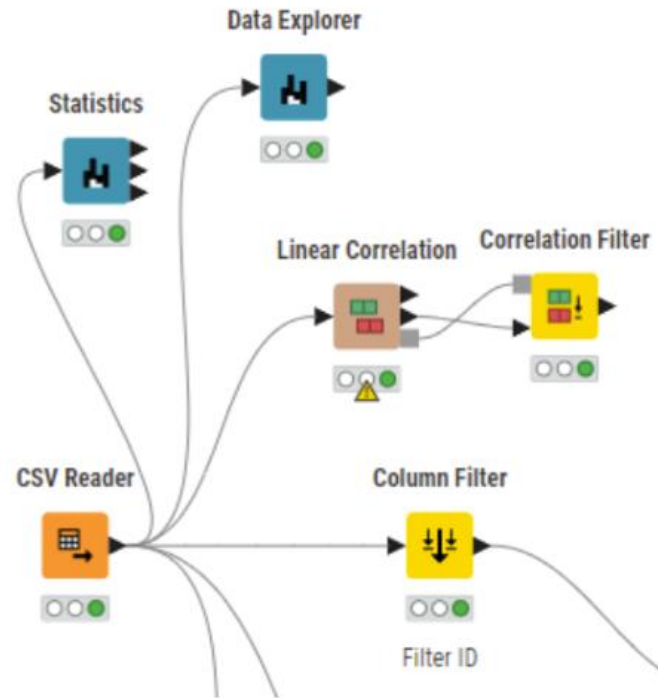
Open the **Column Filter** node's configuration panel. Here, **exclude** the **prod_id** attribute.

This is a **numeric identifier** of each row of the dataset, and adds **no information** to this case.

Column Filter



State of the workflow



Data Preparation - Encoding

Label encoding is a technique used in machine learning to convert **categorical** data into **numerical** form.

Rule Engine



Let's implement this technique for the **review_difficulty** attribute.

This attribute can assume the values:

"Very Easy"

"Easy"

"Average"

"Challenging"

"Very Challenging".

We are going to convert each of them to a **numerical** form.

Data Preparation - Encoding

Open the **Rule Engine** node's **configuration panel**, and, under **Expression**, write the **rules** that will **map the categorical values** to **numerical values**.

Rule Engine



Rule Editor | Flow Variables | Job Manager Selection | Memory Policy

Column List	Category	Description
ROWID	All	
ROWINDEX		
ROWCOUNT		
S ages	Function	
D list_price	? < ?	
I num_reviews	? <= ?	
I piece_count	? = ?	
D play_star_rating	? > ?	
S review_difficulty	? >= ?	
D star_rating	? AND ?	
S theme_name	? IN ?	
D val_star_rating	? LIKE ?	
S country	? MATCHES ?	
	? OR ?	
	? XOR ?	
	FALSE	
	MISSING ?	

Flow Variable List

S knime.workspace

Expression

```

4 // TRUE => "default outcome"
5 $review_difficulty$ = "Very Easy" => 0
6 $review_difficulty$ = "Easy" => 1
7 $review_difficulty$ = "Average" => 2
8 $review_difficulty$ = "Challenging" => 3
9 $review_difficulty$ = "Very Challenging" => 4
  
```

Append Column: Encoder (RD) [I]

Replace Column: S country [v]

Data Preparation - Encoding

Inspect the output in the node monitor.

You should see a **new column, Encoder (RD)**, with the new values.

► 1: Classified values  Flow Variables

Rows: 12261 | Columns: 11

Table Statistics

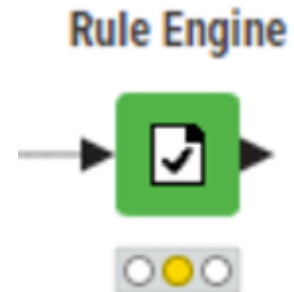
piece_co... Number (inte...	play_star... Number (dou...	review_d... String	star_rating... Number (dou...	theme_n... String	val_star_... Number (dou...	country String	Encoder ... Number (inte...
277	4	Average	4.5	Angry Birdsâ„†	4	US	2
168	4	Easy	5	Angry Birdsâ„†	4	US	1
74	4.3	Easy	4.3	Angry Birdsâ„†	4.1	US	1
1032	3.6	Average	4.6	Architecture	4.3	US	2
744	3.2	Challenging	4.6	Architecture	4.1	US	3
597	3.7	Average	4.9	Architecture	4.4	US	2

Data Preparation – Feature Engineering

Feature engineering is the process of **creating new features** or **modifying** existing ones in a dataset to improve the performance of machine learning models. It involves **selecting, transforming, and creating features** from the raw data to make it more suitable for modeling.

In this case, we will transform all the values corresponding to the **country** acronyms to their corresponding **continent**, e.g, Europe, America, ...

Open the **Rule Engine node's** configuration panel, and, under **Expression**, write the **rules** that will map the **country** acronyms values to their corresponding **continent**.

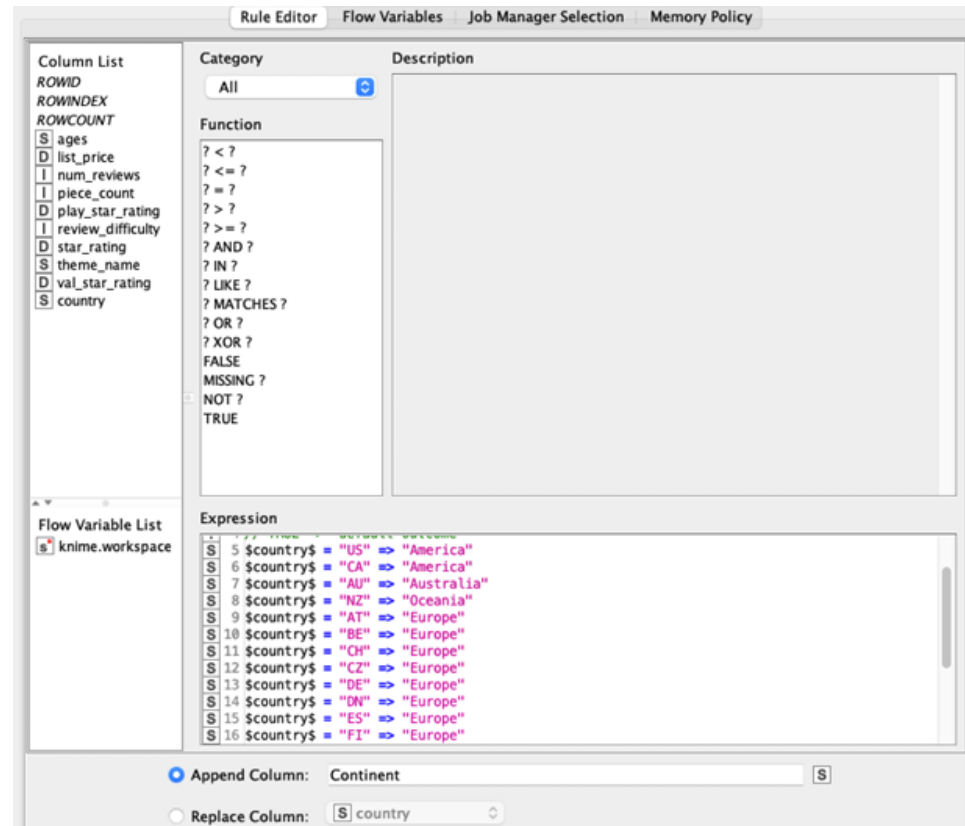


Data Preparation - Encoding

Rule Engine



\$country\$ = "US" → "America"
 \$country\$ = "CA" → "America"
 \$country\$ = "AU" → "Australia"
 \$country\$ = "NZ" → "Oceania"
 \$country\$ = "AT" → "Europe"
 \$country\$ = "BE" → "Europe"
 \$country\$ = "CH" → "Europe"
 \$country\$ = "CZ" → "Europe"
 \$country\$ = "DE" → "Europe"
 \$country\$ = "DN" → "Europe"
 \$country\$ = "ES" → "Europe"
 \$country\$ = "FI" → "Europe"
 \$country\$ = "FR" → "Europe"
 \$country\$ = "GB" → "Europe"
 \$country\$ = "IE" → "Europe"
 \$country\$ = "IT" → "Europe"
 \$country\$ = "LU" → "Europe"
 \$country\$ = "NO" → "Europe"
 \$country\$ = "NL" → "Europe"
 \$country\$ = "PL" → "Europe"
 \$country\$ = "PT" → "Europe"



The screenshot shows the KNIME Rule Editor interface. The top tabs are "Rule Editor", "Flow Variables", "Job Manager Selection", and "Memory Policy". The "Rule Editor" tab is active.

The interface is divided into several sections:

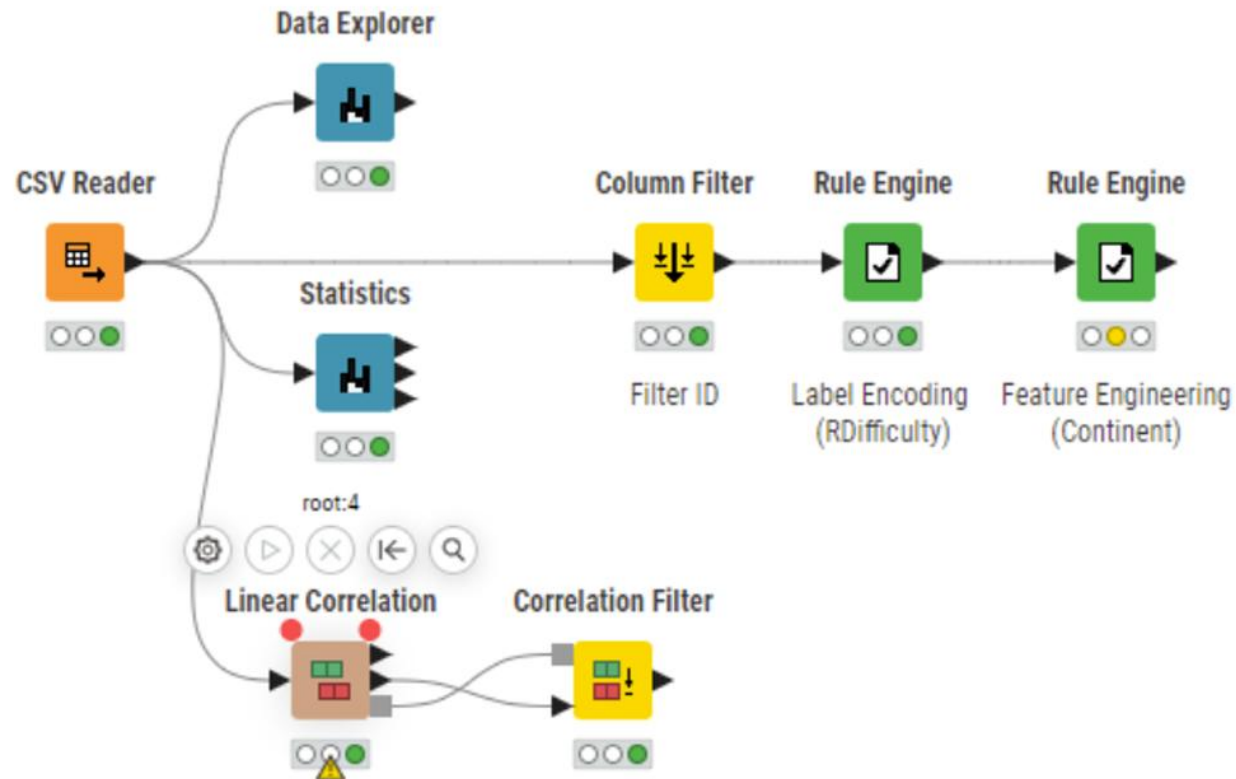
- Column List:** A list of columns including ROWID, ROWINDEX, ROWCOUNT, ages, list_price, num_reviews, piece_count, play_star_rating, review_difficulty, star_rating, theme_name, val_star_rating, and country.
- Category:** A dropdown menu set to "All".
- Description:** A large empty text area for describing the rule.
- Function:** A list of logical functions including "<", "<=", "=", ">", ">=", "AND", "IN", "LIKE", "MATCHES", "OR", "XOR", "FALSE", "MISSING", "NOT", and "TRUE".
- Expression:** A list of expressions showing the mapping of country codes to continent names:

Index	Country	Continent
5	\$country\$ = "US"	"America"
6	\$country\$ = "CA"	"America"
7	\$country\$ = "AU"	"Australia"
8	\$country\$ = "NZ"	"Oceania"
9	\$country\$ = "AT"	"Europe"
10	\$country\$ = "BE"	"Europe"
11	\$country\$ = "CH"	"Europe"
12	\$country\$ = "CZ"	"Europe"
13	\$country\$ = "DE"	"Europe"
14	\$country\$ = "DN"	"Europe"
15	\$country\$ = "ES"	"Europe"
16	\$country\$ = "FI"	"Europe"
- Flow Variable List:** A list of flow variables including knime.workspace.

At the bottom, there are two options for how to handle the column:

- ☒ Append Column: Continent
- ☐ Replace Column: \$country\$

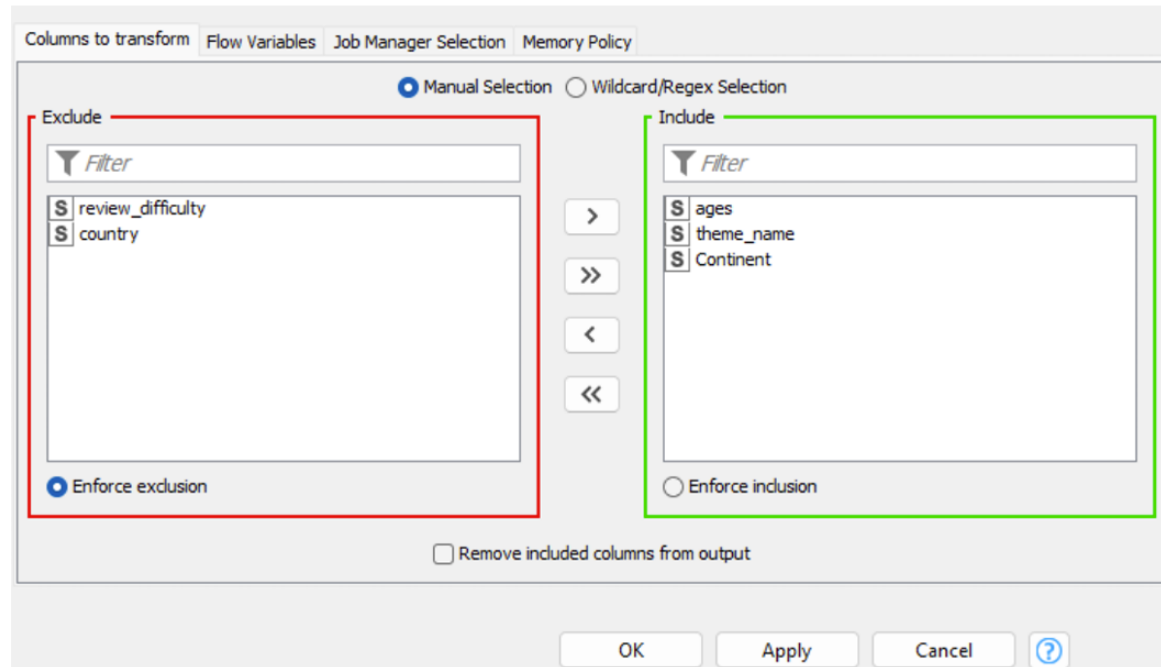
State of the workflow



One-Hot Encoding

In KNIME, a **One to Many** node is a component that allows you to **transform a single input into multiple outputs**.

This is useful when you have a single input that needs to be replicated or distributed to multiple outputs, each of which has a specific condition or transformation.



One-Hot Encoding

You can see new columns, each related to a group of ages, such as 6-12, 12+, etc.

► 1: Processed data  Flow Variables

Rows: 12261 | Columns: 87

Table Statistics

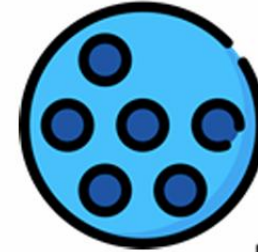
6-12 Number (inte... ▾	12+ Number (inte... ▾	7-12 Number (inte... ▾	10+ Number (inte... ▾	5-12 Number (inte... ▾	8-12 Number (inte... ▾	4-7 Number (inte... ▾
1	0	0	0	0	0	0
1	0	0	0	0	0	0
1	0	0	0	0	0	0
0	1	0	0	0	0	0

Data Preparation – Numeric Outliers

Outlier treatment

Replace

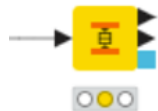
Remove



It depends
mainly on the
number of
outliers.

Numeric Outliers

Numeric Outliers



Outlier Settings | Group Settings | Flow Variables | Job Manager Selection | Memory Policy

Outlier Selection

☒ Manual Selection ☐ Wildcard/Regex Selection

Exclude

Filter

No columns in this list

☒ Enforce exclusion

Include

Filter

- ☒ list_price
- ☒ num_reviews
- ☒ piece_count
- ☒ play_star_rating
- ☒ star_rating
- ☒ val_star_rating
- ☒ Encoder (RD)
- ☒ 6-12
- ☒ 12+
- ☒ 7-12
- ☒ 10+

☐ Enforce inclusion

General Settings

Interquartile range multiplier (k)

Quartile calculation

☐ Use heuristic (memory friendly)

☒ Full data estimate using

☐ Update domain

Outlier Treatment

Apply to

Treatment option

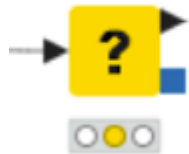
Replacement strategy

Missing Values

Imputation

-**Mean/Median/Mode amputation: replace** missing values with the **mean** (for numerical data), **median** (if the data is skewed) or **mode** (for categorical data)

Missing Value



String

Number (double)

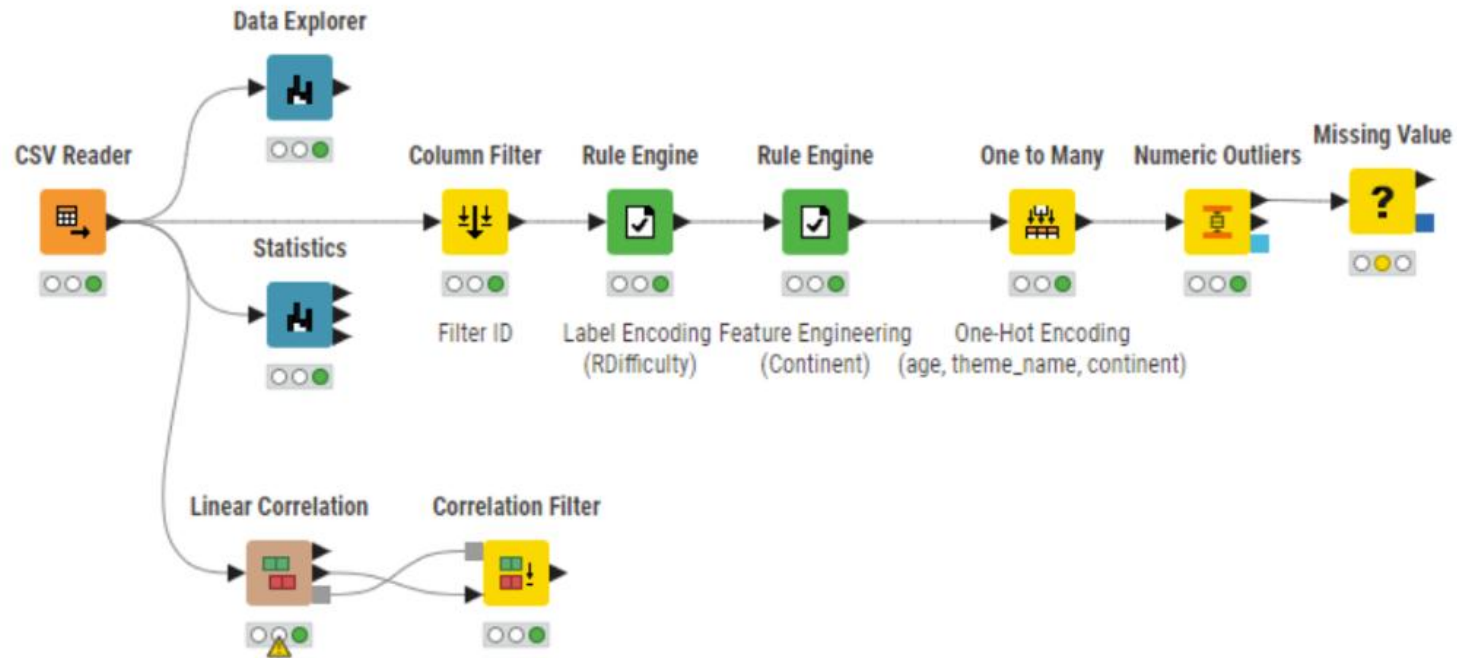
Number (integer)

Most Frequent Value

Mean

Mean

Final Workflow





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